6. Student Projects

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From a Prompt to a Product

An artifact

A self-contained interactive page Claude builds inside the conversation — a dashboard or calculator you use immediately

An app

A web page with a form and a backend that does real work — upload data, get a schedule, a chart, a report

An agent

An app whose menus are replaced by plain language and a model that decides which tools to use

Students describe what they want; Claude writes the code, runs it, and can put it online. To assign or grade any of this, it helps to know what a piece of software actually is.

How Software Works

Frontends and Backends

Runs in a browser

- You open a URL
- The browser runs the **frontend** — HTML, CSS, JavaScript
- Clicks send a request to the **backend**, which does the work and replies

Looks native, but isn’t

- Slack, Spotify, Discord
- Each ships a stripped-down browser inside the app
- Same frontend/backend split underneath

See it yourself

- Right-click any page → “View Page Source”
- That HTML, CSS, and JavaScript is the frontend
- Served by the backend the browser connected to

Running It Locally

127.0.0.1 — “localhost”

- The loopback address: traffic never leaves your machine
- Browser and server both on your own computer
- Where every app starts life before it goes online

Ports

- A number saying which program should receive a request
- A local server often listens on 8000 → visit 127.0.0.1:8000
- Several servers can run at once, each on its own port

When you deploy to the cloud, 127.0.0.1 becomes a real domain name — the app code stays the same.

A Very Small App

```
from fastapi import FastAPI
from fastapi.responses import HTMLResponse

app = FastAPI()

@app.get("/")
def home():
    return HTMLResponse("<h1>Hello, world!</h1>")
```

Serve it (Claude can do this for you), open 127.0.0.1:8000, and the browser sends “GET /”, the server returns the HTML, and you see “Hello, world!” That is the whole shape of a web app — everything else is more of the same.

Artifacts

The Fastest Path to a Working Thing

What you get

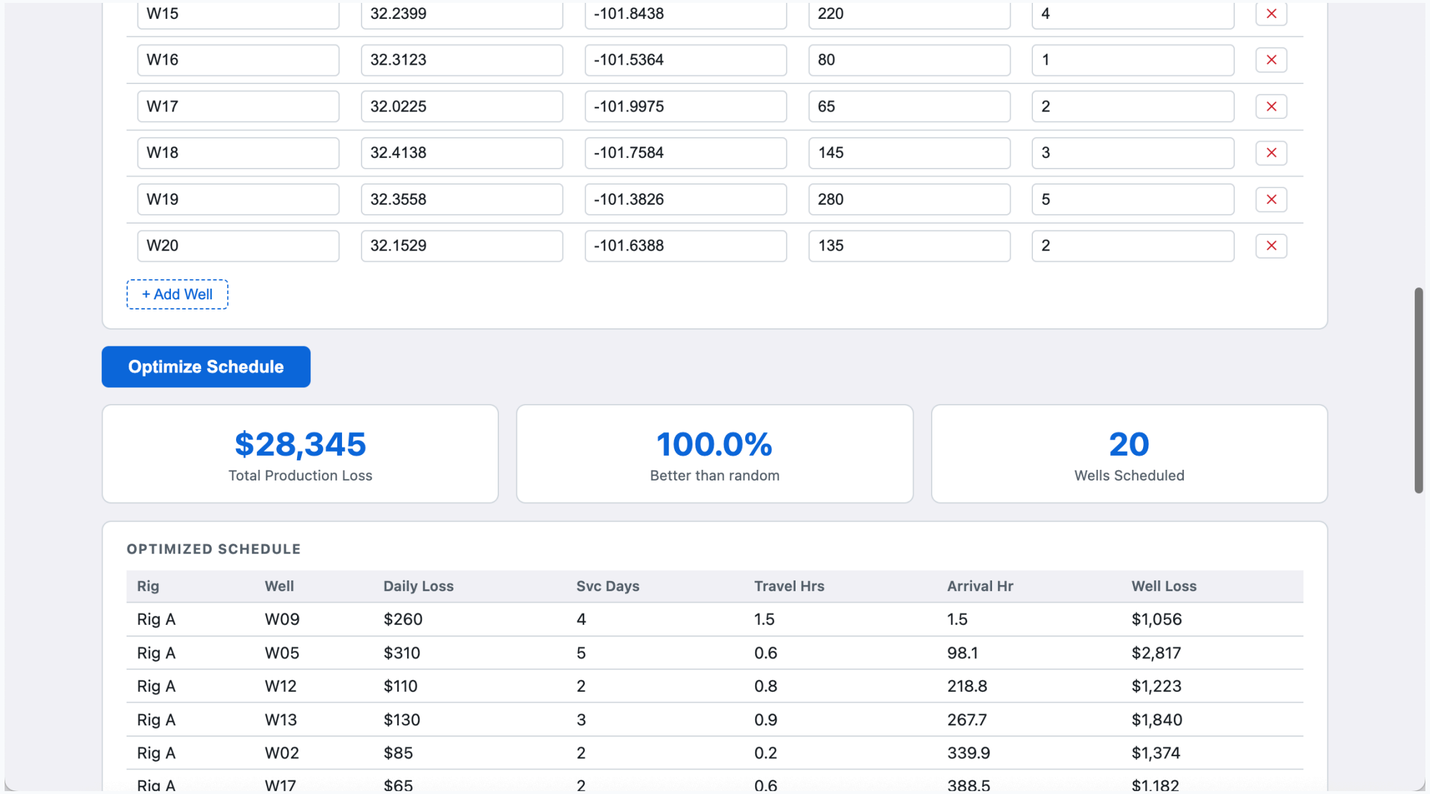
- An interactive page — dashboard, calculator, visualization — built right in the chat
- Use it instantly; ask for changes in the same conversation
- Publish a link, or download it as a single HTML file

When to use it

- The shortest distance from idea to a shareable tool
- No terminal, no server to run
- Great for a one-off tool, a demo, or a student’s first build

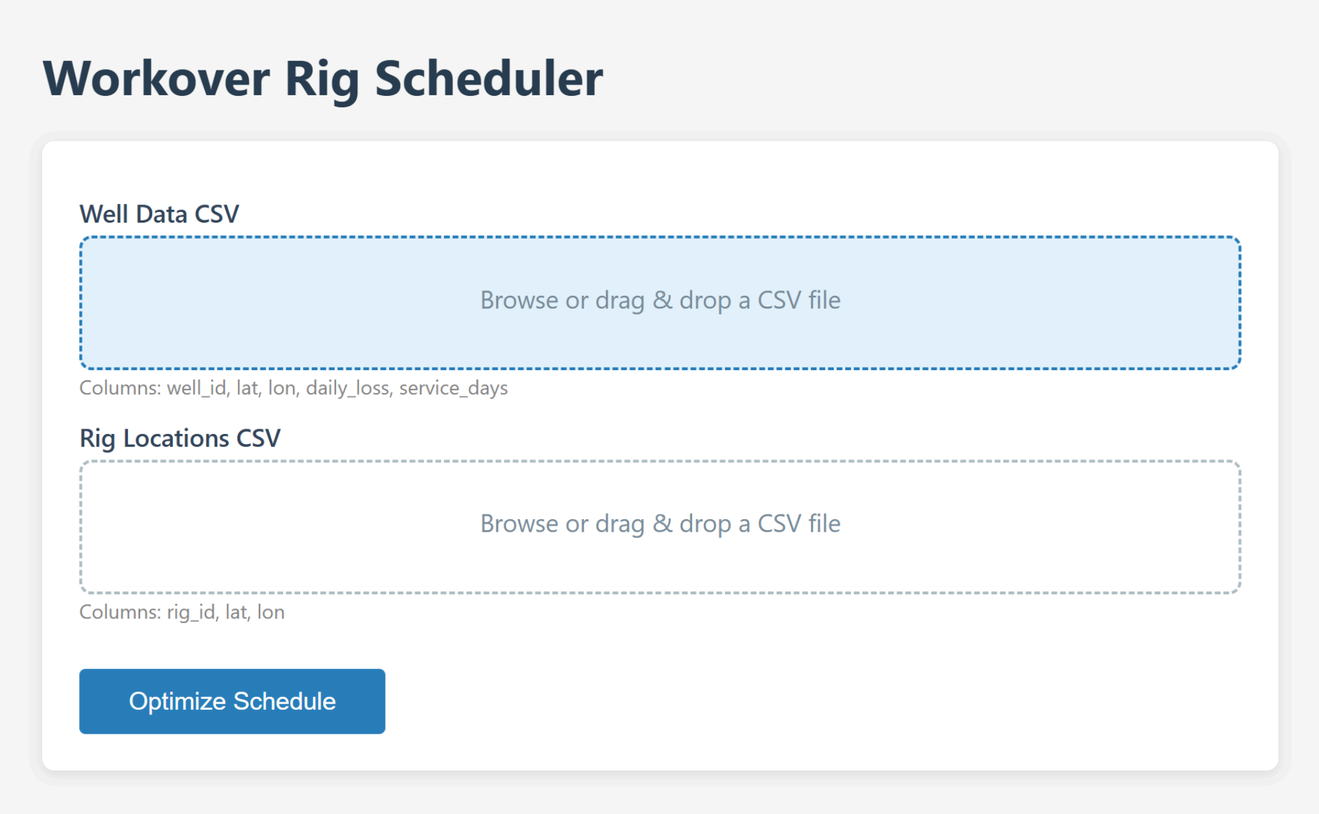
No backend, no deployment — everything runs in the browser. Perfect for a self-contained computation that doesn’t need a database.

Standalone Apps in the Browser



Python runs inside the browser through WebAssembly (Pyodide). The entire app — form, logic, and charts — lives in one HTML file. No server, no install, nothing for the user to set up.

A Real One Students Build



Upload data, click a button, get back a schedule, a route map, and a chart. The backend runs verified Python; the frontend is a simple upload form and a results page. This is a graded deliverable that either works or it doesn’t.

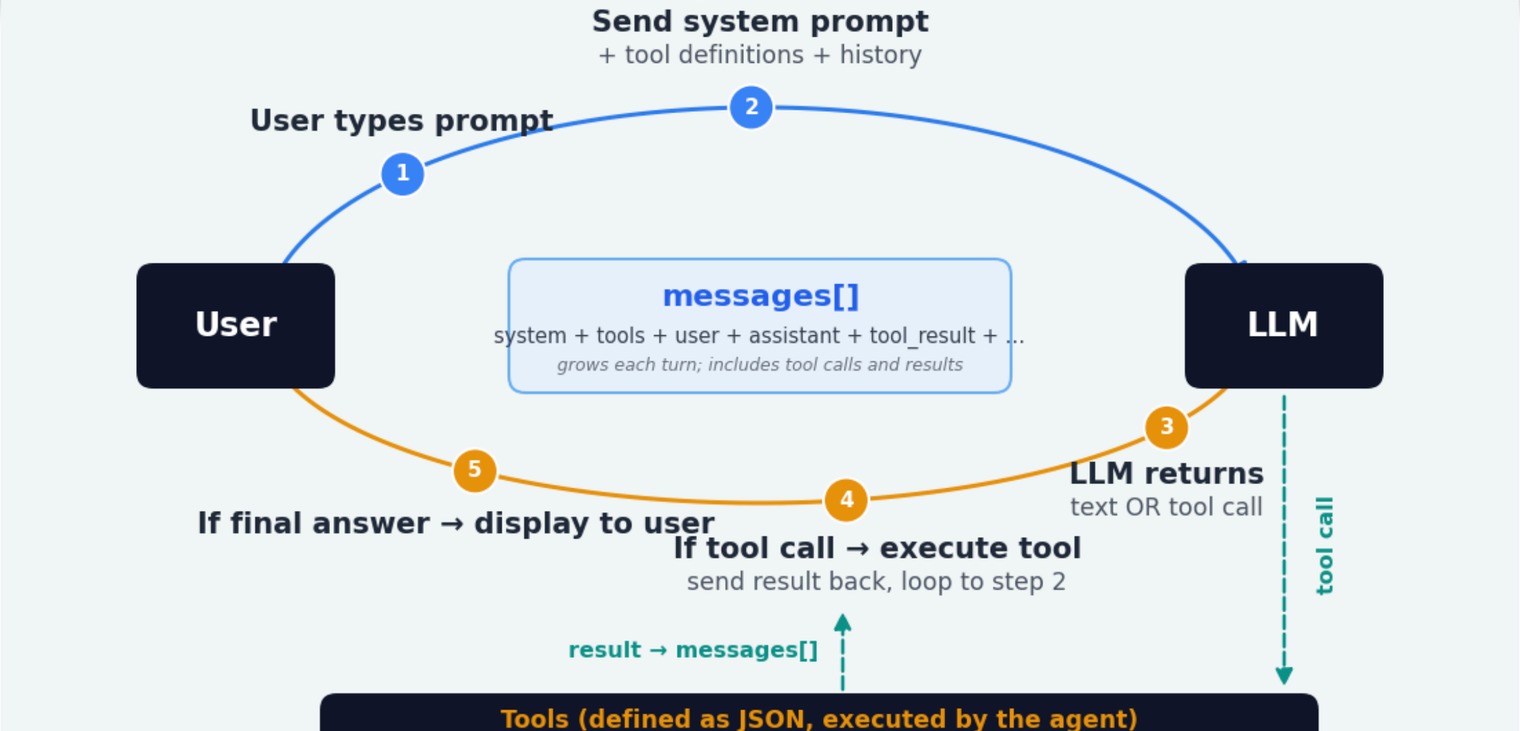
Agents

Chatbot

- User ↔ chatbot ↔ model
- Each turn sends the user's prompt, a system prompt, and the history
- It produces text — nothing more

Agent

- The same loop, plus **tools**
- The model can run code, query a database, call an API
- It decides which tool to use, reads the result, and continues

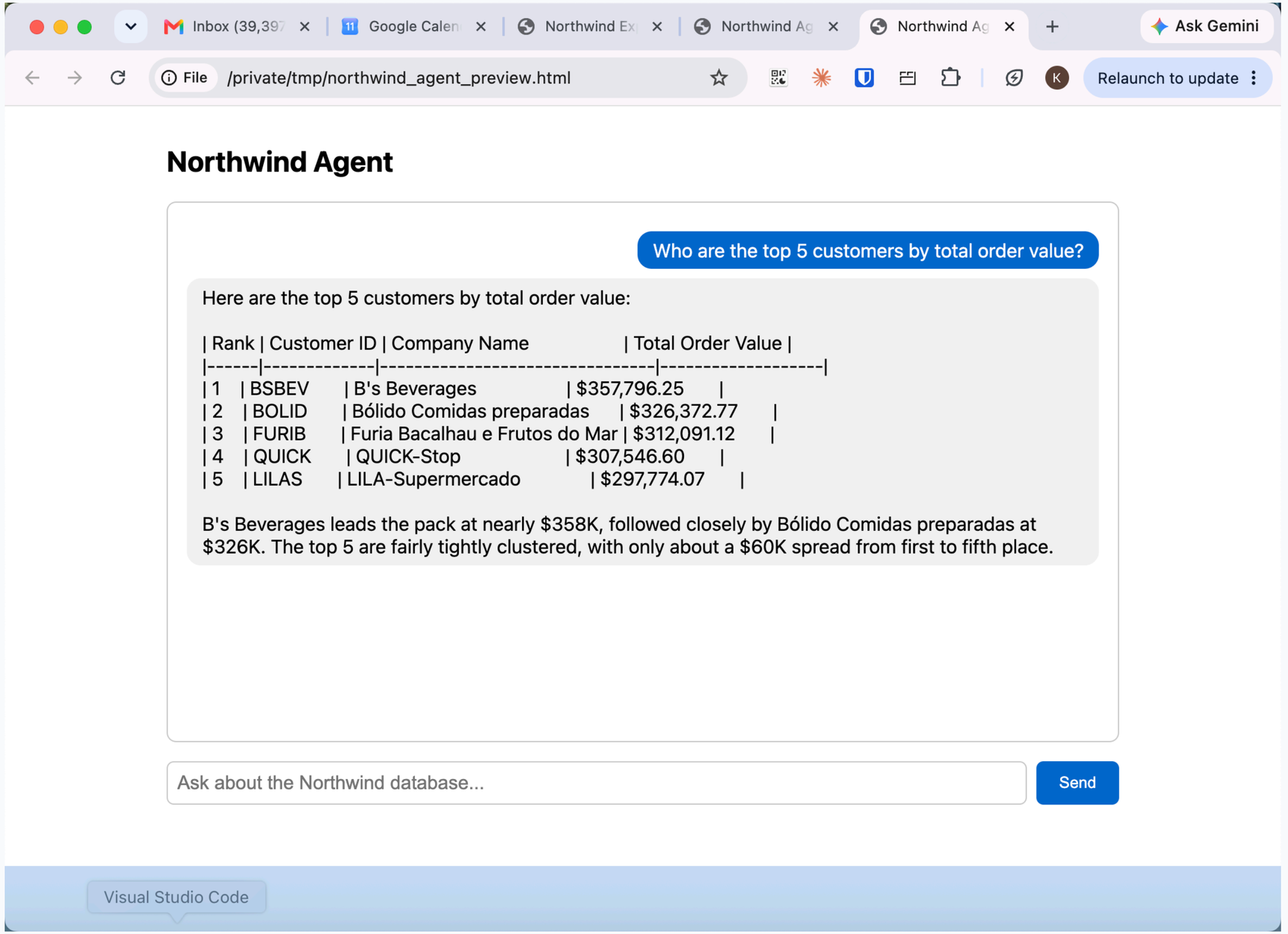


What Makes It an Agent

The designer holds every dial

1. The **system prompt** — the agent’s role, rules, and refusals
2. The **tools** it is given — and, just as important, the ones it is *not*
3. The **permission mode** — what it may do without asking
4. Limits — how many steps, which actions need a human’s yes

An agent is not a different technology. It is a model, a system prompt, and a carefully chosen set of tools — the anatomy from session one, assembled into something that works on its own.



Ask a business question in plain English; the agent writes SQL, runs it against the database, and answers. The

Building It: The Agent SDK

The pieces

- An API key from the Anthropic console, kept in a `.env` file
- The Agent SDK: a few lines to give the model tools and run the loop
- Your tools are just Python functions the model may call

The safety dials

- Allowed tools and disallowed tools — an explicit list
- Permission modes — from “ask me every time” to “run freely”
- Start restrictive; loosen only what you trust

The first and best guardrail is a tool the agent simply does not have. If it cannot send email or delete files, no clever prompt can make it.

Going Live

Two Paths to a Public URL

GitHub Pages — static & Pyodide

- Name the page `index.html`, push, turn on Pages
- Live at a `github.io` address, free, no server
- Good for artifacts and browser-only apps

A PaaS — apps & agents

- Koyeb, Railway, Render go from a GitHub repo to a running service
- They handle the packaging; you skip the server plumbing
- The path for anything with a backend

After you create the account, Claude Code can do the rest — build, configure, and deploy. From a laptop to a public link is a short trip.

Why This Belongs in Your Course

A student who builds an artifact, an app, or an agent has to **decide what it should do** and **judge whether it works.** The building is now easy; the design and the verification are the learning — and the deliverable either runs or it doesn’t.

Breakout

Questions to Discuss

- What tool do you wish existed for your field that a student could build in a week?
- Would it be an artifact, an app, or an agent — and why?
- If students built an agent on real data, what tools would you *withhold* from it?
- What would you look at to grade it — the finished tool, the code, or how they defend its design?